

CHAPTER 8

RAIN LOADS

8.1 DEFINITIONS AND SYMBOLS

8.1.1 Definitions

CONTROLLED DRAINAGE: System intentionally regulating the rate of flow through the primary drains.

PONDING: The accumulation of water caused by the deflection of the roof structure, resulting in added load.

PONDING INSTABILITY: Member instability caused by progressive deflection due to ponding on roofs.

PRIMARY DRAINAGE SYSTEM: Roof drainage system through which water is normally conveyed off the roof.

PRIMARY MEMBERS: For the purposes of determining a susceptible bay, structural members having direct connection to the columns, including girders, beams, and trusses.

SCUPPER: An opening in the side of a building (typically through a parapet wall) for the purpose of draining water off the roof.

SECONDARY DRAINAGE SYSTEM: Roof drainage system at an elevation higher than the primary drainage system, through which water drains off the roof when the primary system is blocked or not working.

SECONDARY MEMBER: For the purposes of determining susceptible bays, structural members not having direct connection to the columns.

SUSCEPTIBLE BAY: A structural bay that is vulnerable to overload from accumulated water.

8.1.2 Symbols

d_h = additional depth of water on the undeflected roof above the inlet of the secondary drainage system at its design flow (i.e., the hydraulic head), in in. (mm)

d_s = depth of water on the undeflected roof up to the inlet of the secondary drainage system when the primary drainage system is blocked (i.e., the static head), in in. (mm)

R = rain load on the undeflected roof, in lb/ft² (kN/m²).
When the phrase “undeflected roof” is used, deflections from loads (including dead loads) shall not be considered when determining the amount of rain on the roof

8.2 ROOF DRAINAGE

Roof drainage systems shall be designed in accordance with the provisions of the code that has jurisdiction. The design flow rate of the secondary (overflow) drains (including roof drains and downstream piping) or scuppers, and their resulting hydraulic head (d_h) shall be based on a rainfall intensity equal to or greater than the 15-min duration/100-year return period (frequency) storm. Primary drainage systems shall be designed for a rainfall intensity equal to or greater than the 60-min duration/ 100-year return period (frequency) storm.

8.3 DESIGN RAIN LOADS

Each portion of a roof shall be designed to sustain the load of all rainwater that will accumulate on it if the primary drainage system for that portion is blocked plus the uniform load caused by water that rises above the inlet of the secondary drainage system at its design flow.

$$R = 5.2(d_s + d_h) \quad (8.3-1)$$

$$R = 0.0098(d_s + d_h) \quad (8.3-1.si)$$

If the secondary drainage systems contain drain lines, such lines and their point of discharge shall be separate from the primary drain lines. Rain loads shall be based on the total head (static head [d_s] plus hydraulic head [d_h]) associated with the design flow rate for the specified secondary drains and drainage system. The total head corresponding to the design flow rate for the specified drains shall be based on hydraulic test data.

8.4 PONDING INSTABILITY AND PONDING LOAD

Susceptible bays shall be investigated by structural analysis to ensure that they possess adequate stiffness to preclude progressive deflection (i.e., instability) and adequate strength to resist the additional ponding load. Any of the following conditions shall be deemed to create susceptible bays: (1) bays with a roof slope less than 1/4 in. per foot (1.19°) when the secondary members are perpendicular to the free draining edge, (2) bays with a roof slope less than 1 in. per foot (4.76°) when the secondary members are parallel to the free draining edge, (3) bays with a roof slope of 1 in. per foot (4.76°) and a span to spacing ratio for the secondary members greater than 16 when the secondary members are parallel to the free draining edge, or (4) bays on which water accumulates (in whole or in part) when the primary drain system is blocked but the secondary drain system is functional. The larger of the snow load or the rain load equal to the design condition for a blocked primary drain system shall be used in this analysis.

8.5 CONTROLLED DRAINAGE

Roofs equipped with hardware on the primary drainage system designed to intentionally regulate the rate of drainage shall be equipped with a secondary drainage system at a higher elevation. Controlled flow roof drains shall not be used on the secondary drainage system.

8.6 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

No consensus standards and other documents that shall be considered part of this standard are referenced in this chapter.

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